Time Series and Fourier Analysis

EES 4891/5891

Probability & Statistics for Geosciences

Jonathan Gilligan

Class #25: Tuesday, April 08 2025

Learning Goals

Learning Goals

- Learn what makes time series data different from other kinds of data.
- Learn what autocorrelation is, and why it matters.
- Focus on periodic variations:
  - Sources of periodic signals in environmental data
  - Fourier’s Theorem
  - Fourier Transforms
  - POwer spectra and spectral power density
- Doing Fourier Transforms in R with Fast Fourier Transforms (FFT)
- Learn what aliasing is
  - Why it’s a problem
  - How to avoid it

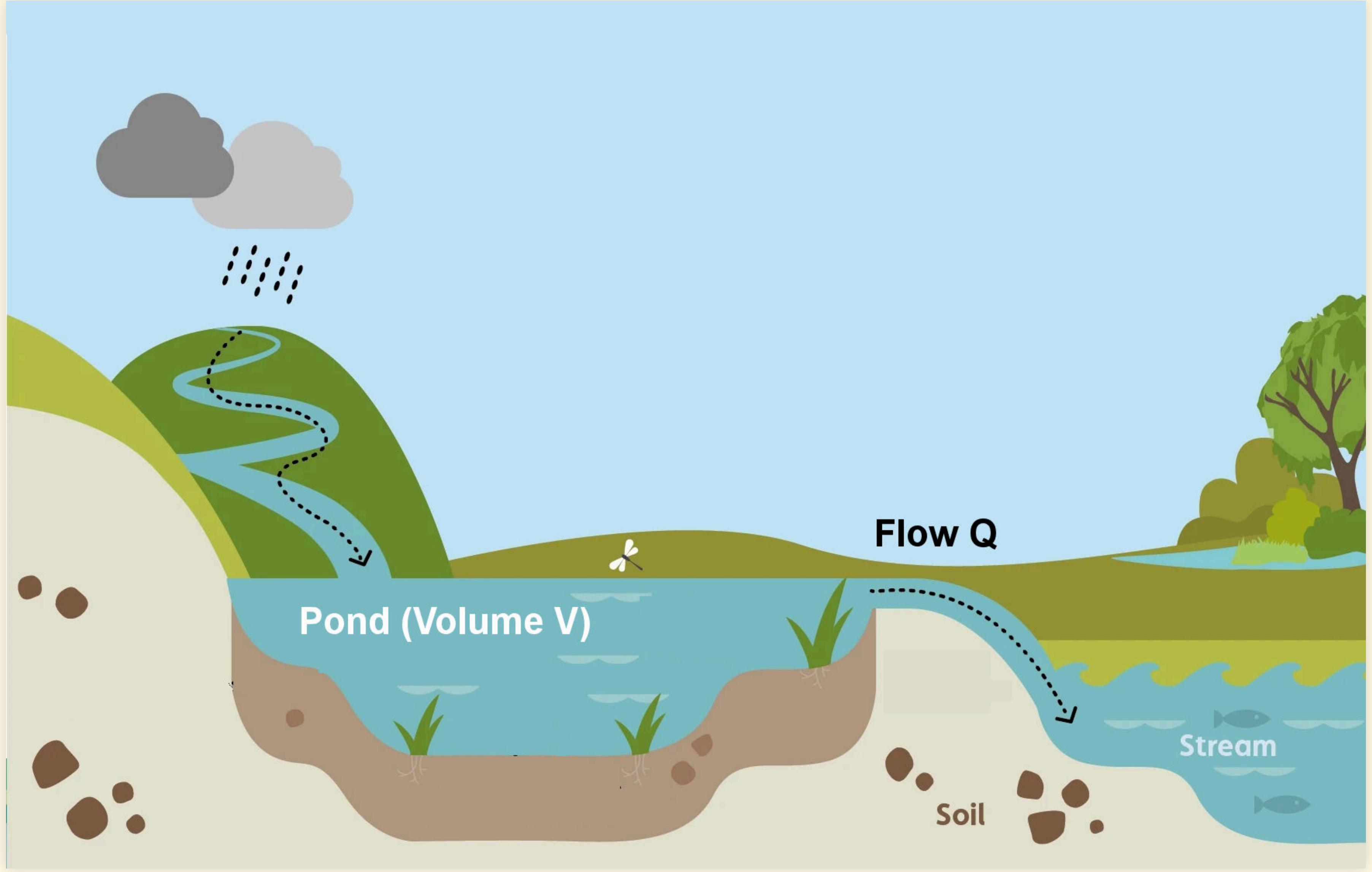
Time Series

Time Series

- Time Series:
  - Observations that are ordered in time
    - We will only consider data that are evenly spaced
    - There are advanced methods for irregularly spaced data, but we won’t study them in this course
- Why are time series different?
  - Causal relationships
    - Earlier events may influence later ones
- Independent, identically distributed (iid) data
  - Each observation is independent of all the others
  - It doesn’t matter in what order they occur
- Time series:
  - Observations may not be independent
  - Later observations may correlated with earlier ones

Example

- Example:
  - Concentration of nitrate in a pond of volume V
    - Fed by one stream, drained by another:
      - Volume Q exchanged every day



- Time zero, a rainstorm washes mass M of nitrate into the pond
  - Initial concentration M/V
  - 1 day later

$$\frac{M}{V}\left(\frac{V-Q}{V}\right)=\frac{M}{V}\left(1-\frac{Q}{V}\right)$$

  - 2 days later:

$$\frac{M}{V}\left(1-\frac{Q}{V}\right)^2$$

  - 3 days later:

$$\frac{M}{V}\left(1-\frac{Q}{V}\right)^3$$

More Complicated Example

- What does a “100-year flood” mean?
  - A probability of 0.01 that there will be a flood in any year.
- If there is a 100-year flood this year, what is the probability that there will be another 100-year flood next year?
  - Most people will say 0.01
  - In fact, a 100-year flood is slightly more likely if there was a big flood the year before
  - And slightly less likely if there wasn’t a flood the year before.

VOL. 4, NO. 5 WATER RESOURCES RESEARCH OCTOBER 1968

Noah, Joseph, and Operational Hydrology

BENOIT B. MANDELBROT

JAMES R. WALLIS

International Business Machines Research Center
Yorktown Heights, New York 10598

- Autocorrelated noise
  - Regular regression: $y_j = f(x_j) + \epsilon_j$,
    - ϵ_j are iid: no correlation
  - Time-series: ϵ_{j+1} may be correlated with ϵ_j
    - Autocorrelation
- Wet years follow wet years
- Dry years follow dry years

Periodic Variations

- Many environmental time series show periodic variations:
  - Hourly data show diurnal cycles
  - Daily or monthly data show annual cycles
  - Brightness of the sun has an 11-year cycle
  - Ice-ages have Milankovitch cycles (41 kyear, 100 kyear, etc.)
- Time-frequency duality
  - Any time-series x_j, with N observations sampled at intervals τ can be expressed as a sum of N periodic functions with frequencies from $1/\tau$ to N/τ.

- Time-series: $x_j, j = 1, \ldots, N$,
  - Sampled at intervals τ: $t_j = j\tau$
- Periodic representation (Fourier Theorem):

$$x_j = \frac{a_0}{2} + \sum_{k=1}^{N} a_k \cos(k\omega t_j) + b_k \sin(k\omega t_j)$$

$$\omega = \frac{2\pi}{N\tau}$$

$$a_k = \frac{2}{N} \sum_{j=1}^{N} x_j \cos(k\omega t_j)$$

$$b_k = \frac{2}{N} \sum_{j=1}^{N} x_j \sin(k\omega t_j)$$

Complex Numbers

- Complex number $z = u + iv$, where $i = \sqrt{-1}$

$$e^{ix} = \cos(x) + i\sin(x)$$

- Fourier Theorem with complex numbers:

$$x_j = \sum_{k=-N}^{N} c_k e^{ik\omega t_j}$$

$$\omega = \frac{2\pi}{N\tau}$$

$$c_k = \begin{cases} \frac{1}{2}(a_k - ib_k) & k \geq 0 \\ \frac{1}{2}(a_k + ib_k) & k < 0 \end{cases}$$

Fourier Transform

- Continuous function $f(t)$, t varies continuously from $-\infty$ to $+\infty$:
  - Fourier transform $g(\omega)$:

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} g(\omega)e^{+i\omega t}\mathrm{d}\omega$$

$$g(\omega) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t}\mathrm{d}t$$

- Discrete observations: x_j sampled at N evenly-spaced intervals $t_i = i\tau$
  - Discrete Fourier transform z_k:

$$x_j = \frac{1}{N}\sum_{k=0}^{N-1} z_k e^{+ik\omega t_j}$$

$$z_k = \sum_{j=1}^{N} x_j e^{-ik\omega t_j}$$

$$\omega = \frac{2\pi}{N\tau}$$

Power Spectrum

- Parseval's Theorem:
  - Energy in a signal $f(t)$

$$E = \int_{-\infty}^{+\infty} |f(t)|^2 \mathrm{d}t$$
$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} |g(\omega)|^2 \mathrm{d}\omega$$

  - Power spectral density

$$S(\omega) = |g(\omega)|^2$$

    - Energy per interval of frequency

- Discrete Parsival's Theorem:
  - Energy in a signal x_j

$$E = \sum_{j=1}^{N} |x_j|^2$$
$$= \frac{1}{N} \sum_{k=0}^{N-1} |z_k|^2$$

  - Power spectral density

$$S(k\omega) = |z_k|^2$$
$$\omega = \frac{2\pi}{N\tau}$$

    - Energy per unit of frequency

Fourier Transforms in R

Fourier Transforms in R

- `fft()`: Fast Fourier Transform

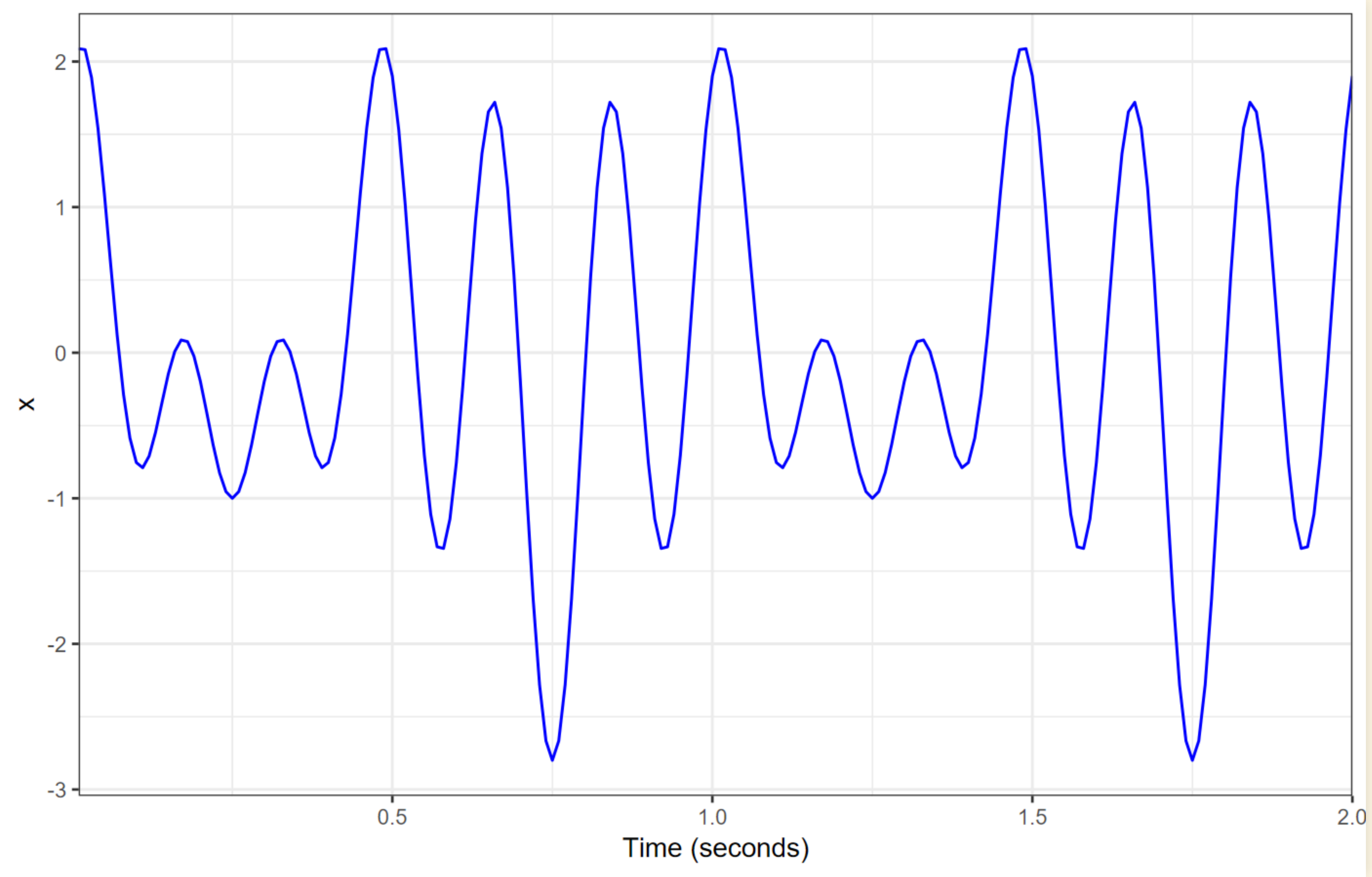
```
df <- tibble(
  i = seq(200),
  t = 2 * i / length(i),
  x = 0.7 * cos(2 * pi * t * 2) +
      1.2 * cos(2 * pi * t * 6) +
      0.9 * sin(2 * pi * t * 5)
  )

df_ft <- tibble(z = fft(df$x),
                i = seq_along(z),
                f = i - 1) |>
  mutate(ps = Mod(z * Conj(z)) / n(),
         f = ifelse(f > n()/2, f - n(), f) / max(df$t))
        |>
  relocate(i, f, z, ps)

head(df_ft)
```

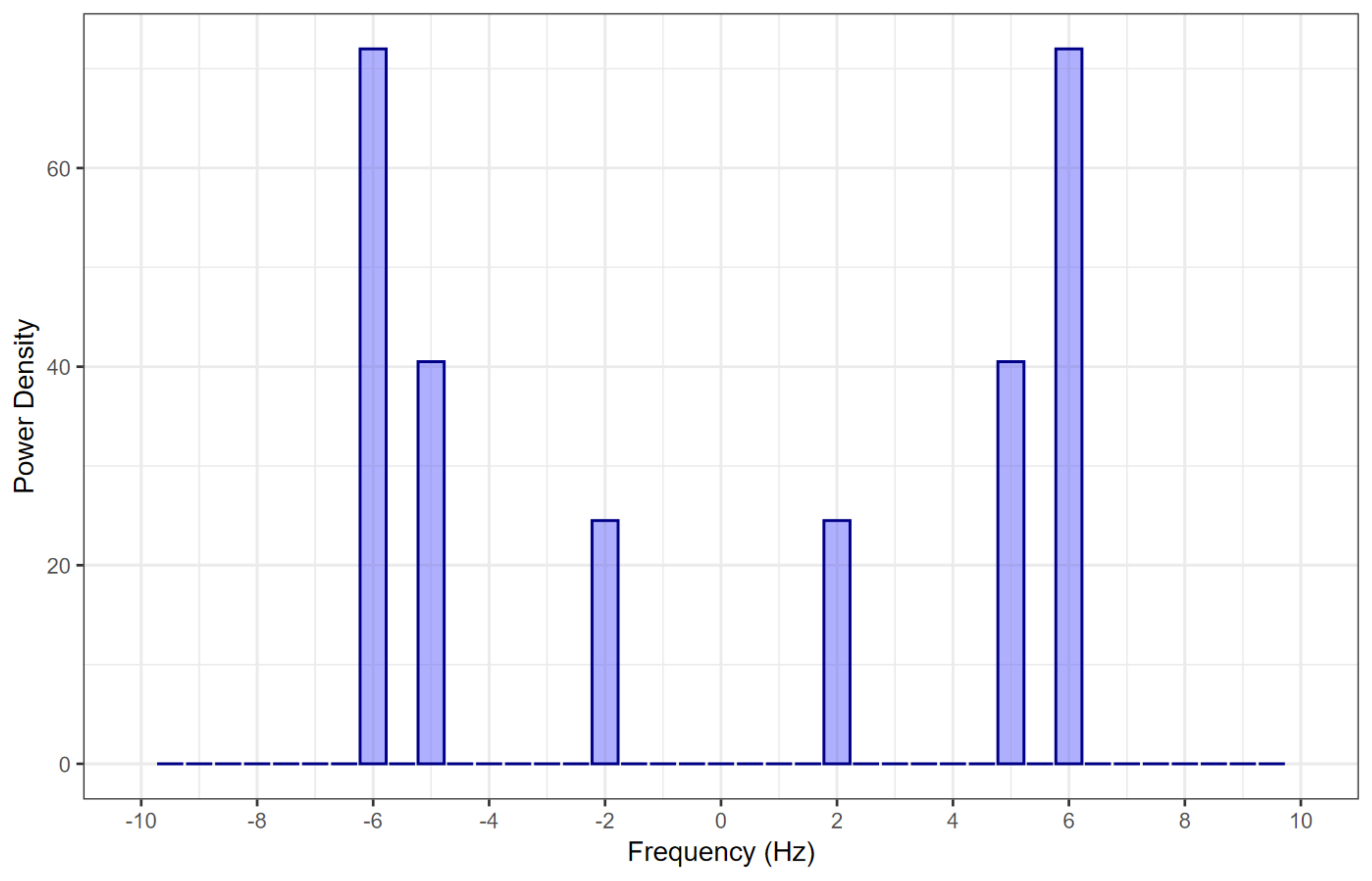
```
## # A tibble: 6 × 4
##       i     f z                                  ps
##   <int> <dbl> <cpl>                           <dbl>
## 1     1   0   -6.128431e-14+0.000000e+00i 1.88e-29
## 2     2   0.5  2.409184e-14+1.942890e-15i 2.92e-30
## 3     3   1   -6.883383e-15+3.974598e-14i 8.14e-30
## 4     4   1.5 -2.495226e-14-6.661338e-15i 3.33e-30
## 5     5   2    6.944803e+01+8.773326e+00i 2.45e+ 1
## 6     6   2.5  3.367053e-14-1.406289e-15i 5.68e-30
```

- Note: We can interpret the higher frequencies ($k > N/2$) as negative frequencies ($k \to k - N$)

$$
\begin{aligned}
z_k &= \sum_{j=1}^{N} x_j e^{-ik\omega t_j} \\
\omega &= \frac{2\pi}{N\tau} \\
k\omega t_j &= k\frac{2\pi}{N\tau} j\tau = 2\pi jk/N \\
e^{i(k+N)\omega t_j} &= e^{i2\pi j(k+N)/N} = e^{i2\pi j(k/N+1)} \\
e^{i2\pi(v+1)} &= e^{i2\pi v}, \\
&\text{so} \\
z_{k\pm N} &= z_k
\end{aligned}
$$

Plotting Data and Fourier Transform






```
summarize(df, power = sum(x^2))
```

```
## # A tibble: 1 × 1
##   power
##   <dbl>
## 1   274
```

```
summarize(df_ft, power = sum(ps))
```

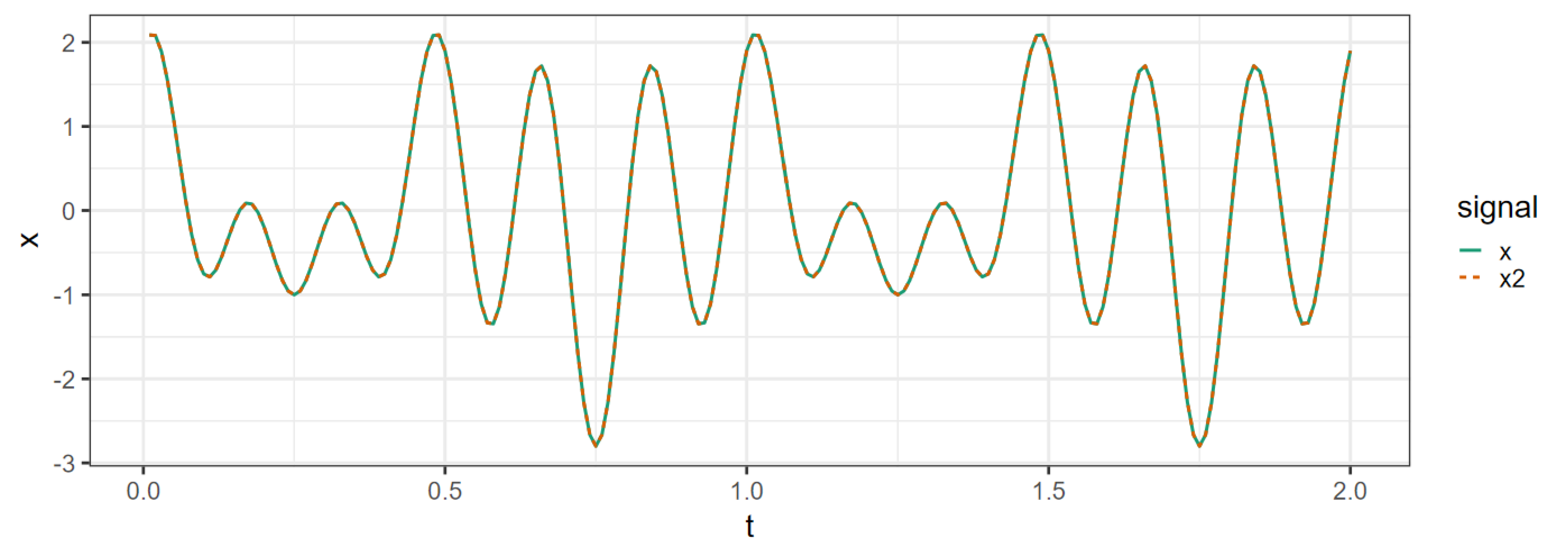
```
## # A tibble: 1 × 1
##   power
##   <dbl>
## 1   274
```

Inverse Fourier Transform

```
df2 <- df |> mutate(x2 = Re(fft(df_ft$z, inverse = TRUE)) / n())
df2 |> summarize(rmse = sqrt(mean((x - x2)^2)))
```

```
## # A tibble: 1 × 1
##       rmse
##      <dbl>
## 1 4.12e-16
```

```
df2 |> pivot_longer(c(x, x2), names_to = "signal", values_to = "x") |>
ggplot(aes(x = t, y = x, color = signal, linetype = signal)) +
  scale_color_brewer(palette = "Dark2") +
  geom_line(linewidth = 1)
```



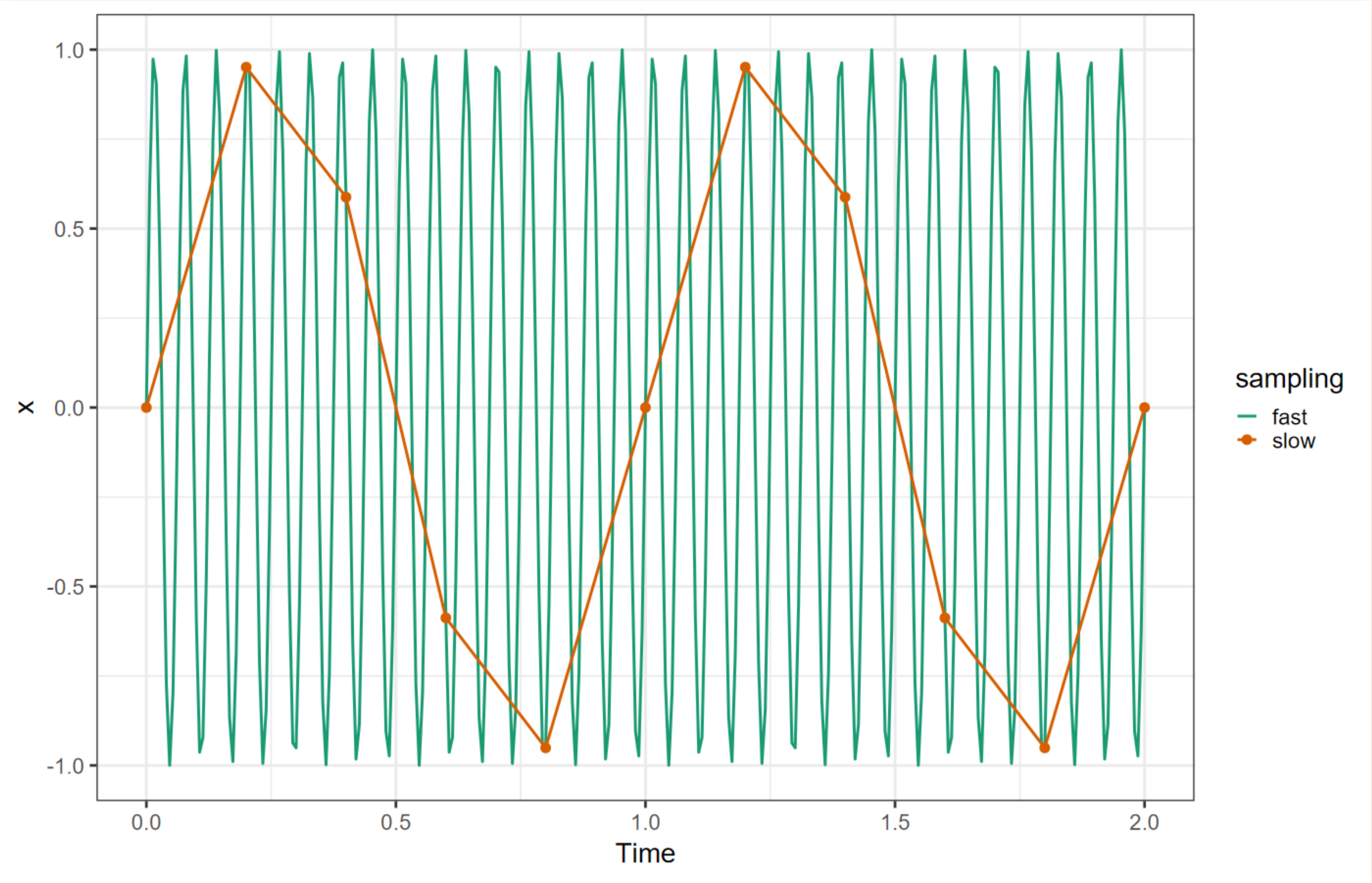
Sampling

Sampling

- Remember that $z_k = z_{k \pm N}$
  - This creates a problem called “aliasing”
- If we sample N points with time interval τ, $\omega = 2\pi/(N\tau)$
  - z_0 is the coefficient for zero-frequency, $0 \times \omega$
  - z_N is the coefficient for $N\omega = 2\pi/\tau$, which means one cycle per period τ.
  - But $z_0 = z_{0+N}$, so our discrete Fourier transform can’t tell the difference between constant data and data that’s oscillating with period τ.
- All frequencies between $k = N/2$ and $k = N$ will appear like negative frequencies between $-N/2$ and 0.
- The highest frequency we can reliably measure is $k = N/2$, which has period 2τ
- When we sample with interval τ, any signal with frequency $f > 1/(2\tau)$ will be “aliased” and will appear in our sampled data with a frequency between $-1/(2\tau)$ and $1/(2\tau)$

Example of Aliasing

```r
tau = 0.2
omega = 2 * pi * 3.2 / tau

fast <- tibble(time = seq(0, 10 * tau, tau / 30),
               x = sin(time * omega),
               sampling = "fast")

slow <- tibble(time = seq(0, 10 * tau, tau),
               x = sin(time * omega),
               sampling = "slow")

df <- bind_rows(fast, slow)

ggplot(df, aes(x = time, y = x, color = sampling)) +
  geom_line(linewidth = 1) +
  geom_point(data = slow, size = 3) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Time", y = "x")
```



- Rule: Make sure that you sample at least twice as fast as the highesst frequency that appears in the signal.
  - With electronic instruments, use electronic filters to remove frequencies higher than half your sampling frequency.
  - Good data loggers will do this automatically.